

Vieta's theorem and factorising a quadratic trinomial

The sum and the product of the roots, read straight off the coefficients — a check, a shortcut and a factorising method in one.

LESSON 68–69

2 × 40 MIN

ALGEBRA 8, CHAPTER III §23

STAGE 9 · 2.4

LESSON CLOCK



Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

LEARNING OBJECTIVES

- State and use Vieta's theorem.
- Check a pair of roots without substituting.
- Find roots by inspection for a reduced quadratic.
- Factorise $ax^2 + bx + c$ using its roots.

TEXTBOOK

National Chapter III, pp. 141–146

Cambridge Learner's Book pp. 36–39

Workbook Workbook 2.4

01 Explanation

The theorem

VIETA'S THEOREM

If x_1 and x_2 are the roots of $ax^2 + bx + c = 0$, then

$$x_1 + x_2 = -\frac{b}{a} \quad \text{and} \quad x_1 \cdot x_2 = \frac{c}{a}$$

For a **reduced** equation $x^2 + px + q = 0$ (with $a = 1$) it is even simpler:

$$x_1 + x_2 = -p \quad \text{and} \quad x_1 \cdot x_2 = q$$

Check it on $x^2 - 5x + 6 = 0$, whose roots are 2 and 3: $2 + 3 = 5 = -(-5) \checkmark$ and $2 \cdot 3 = 6 \checkmark$.

Three uses

1 · As a check. After solving, add and multiply your two roots. If they do not give $-\frac{b}{a}$ and $\frac{c}{a}$, go back — this takes five seconds and catches almost every arithmetic slip.

2 · To find roots by inspection. For $x^2 - 7x + 12 = 0$, look for two numbers with sum 7 and product 12: 3 and 4. No formula needed.

3 · To find an unknown coefficient. If one root of $x^2 + bx - 10 = 0$ is 5, then the product of the roots is -10 , so the other root is -2 , and the sum $5 + (-2) = 3 = -b$ gives $b = -3$.

Factorising the trinomial

THEOREM

If x_1 and x_2 are the roots of $ax^2 + bx + c = 0$, then

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

So factorising a quadratic trinomial and solving the equation are the same job. Find the roots, then write the brackets:

$$2x^2 - 7x + 3: \text{ roots } 3 \text{ and } 0.5, \text{ so } = 2(x - 3)(x - 0.5) = (x - 3)(2x - 1)$$

DO NOT FORGET THE A

$2x^2 - 7x + 3$ is not $(x - 3)(x - 0.5)$ — that expands to $x^2 - 3.5x + 1.5$. The leading coefficient must be carried through.

If $D < 0$ there are no roots, and the trinomial **cannot** be factorised over the real numbers.

02 Worked examples

EXAMPLE 1 Solve $x^2 - 9x + 20 = 0$ by inspection.

- 1 Sum of roots = 9, product = 20.
Vieta for a reduced equation.
- 2 Two numbers with product 20: 1·20, 2·10, 4·5.
- 3 $4 + 5 = 9$ ✓
- 4 $x = 4$ and $x = 5$

ANSWER

$$x = 4, x = 5$$

EXAMPLE 2 One root of $x^2 + bx - 21 = 0$ is 7. Find the other root and b .

- 1 $x_1 \cdot x_2 = -21$
The product equals the free term.
- 2 $7 \cdot x_2 = -21$, so $x_2 = -3$
- 3 $x_1 + x_2 = 7 + (-3) = 4$
- 4 $-b = 4$, so $b = -4$

ANSWER

$$x_2 = -3, b = -4$$

EXAMPLE 3 *Factorise $3x^2 - 5x - 2$*

$$① \quad D = 25 + 24 = 49$$

$$② \quad x = \frac{5 \pm 7}{6}$$

The formula.

$$③ \quad x_1 = 2, x_2 = -\frac{1}{3}$$

$$④ \quad 3(x - 2)\left(x + \frac{1}{3}\right) = (x - 2)(3x + 1)$$

Multiply the 3 into the second bracket.

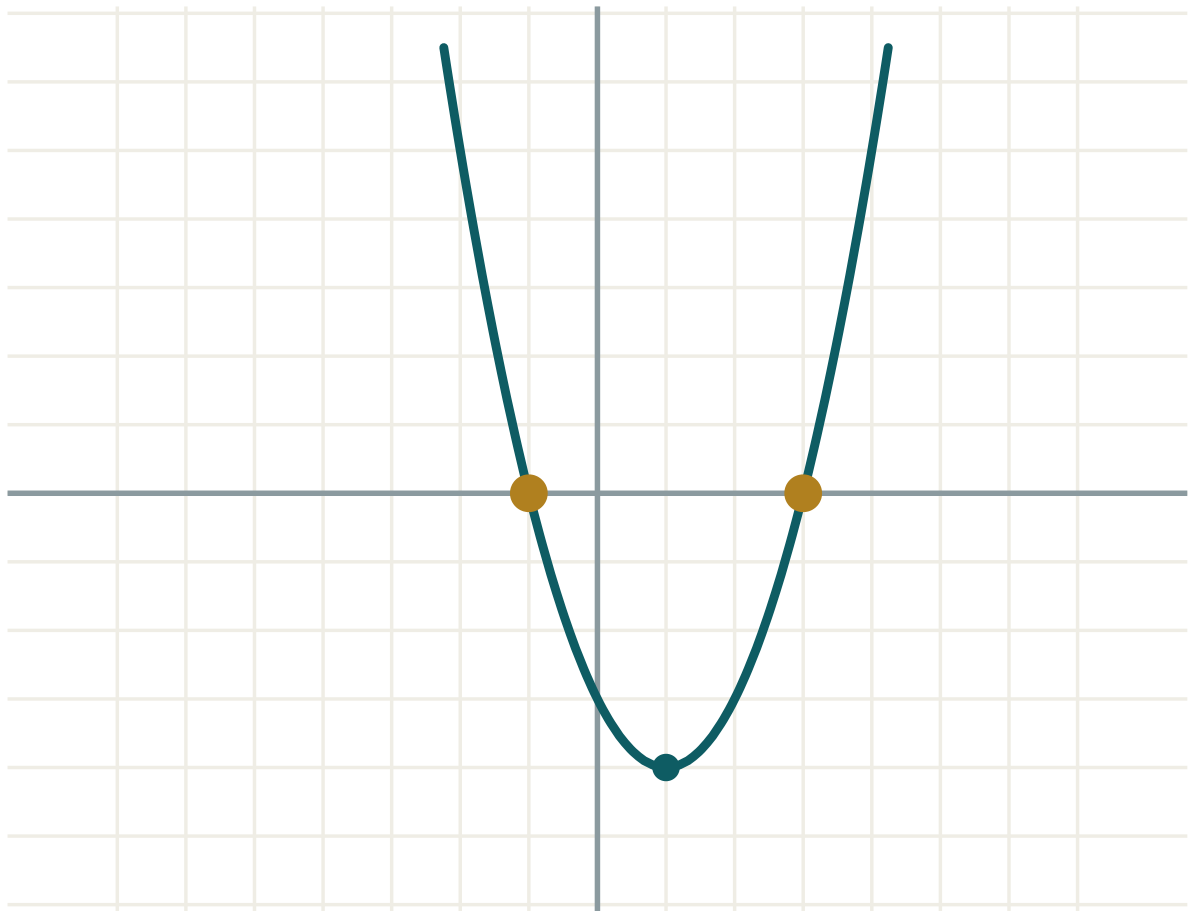
ANSWER

$$(x - 2)(3x + 1)$$

03 **Interactive model**

Read $x_1 + x_2$ and $x_1 \cdot x_2$ from the model and compare with $-b/a$ and c/a .

Sum and product of the roots



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Vieta's theorem	Viyet teoremasi	Теорема Виета
Sum of the roots	Ildizlar yig'indisi	Сумма корней
Product of the roots	Ildizlar ko'paytmasi	Произведение корней
Reduced quadratic equation	Keltirilgan kvadrat tenglama	Приведённое квадратное уравнение
Quadratic trinomial	Kvadrat uchhad	Квадратный трёхчлен
By inspection	Tanlash yo'li bilan	Подбором
Linear factors	Chiziqli ko'paytuvchilar	Линейные множители
Converse theorem	Teskari teorema	Обратная теорема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $x^2 - 7x + 12 = 0$, the sum of the roots is:

For the same equation, the product is:

For $2x^2 + 6x - 8 = 0$, the sum of the roots is:

$x^2 - 5x + 6$ factorises to:

$(x - 2)(x - 3)$

$(x + 2)(x + 3)$

$(x - 1)(x - 6)$

it does not factorise

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the sum of the roots of $x^2 - 4x + 3 = 0$

ANSWER 4

2. Find the product of the roots of $x^2 - 4x + 3 = 0$

ANSWER 3

3. Solve $x^2 - 5x + 6 = 0$ by inspection

ANSWER $x = 2, x = 3$

4. Solve $x^2 - 7x + 12 = 0$ by inspection

ANSWER $x = 3, x = 4$

5. Solve $x^2 + 5x + 6 = 0$ by inspection

ANSWER $x = -2, x = -3$

6. Factorise $x^2 - 3x + 2$

ANSWER $(x - 1)(x - 2)$

7. Factorise $x^2 - 16$

ANSWER $(x - 4)(x + 4)$

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 9x + 20 = 0$ by inspection

ANSWER $x = 4, x = 5$

2. Solve $x^2 + x - 12 = 0$ by inspection

ANSWER $x = 3, x = -4$

3. One root of $x^2 + bx - 21 = 0$ is 7. Find the other root and b .

ANSWER $x_2 = -3, b = -4$

4. For $2x^2 + 6x - 8 = 0$, find the sum and product of the roots.

ANSWER -3 and -4

5. Factorise $x^2 - 2x - 15$

ANSWER $(x - 5)(x + 3)$

6. Factorise $3x^2 - 5x - 2$

ANSWER $(x - 2)(3x + 1)$

7. Write a reduced quadratic whose roots are 6 and -2 .

ANSWER $x^2 - 4x - 12 = 0$

Hard · stretch and reason

7 PROBLEMS

1. The roots of $x^2 + px + q = 0$ are 3 and -5 . Find p and q .

ANSWER $p = 2, q = -15$

2. One root of $2x^2 - 9x + c = 0$ is 4. Find c and the other root.

ANSWER Sum = 4.5, so the other root is 0.5 and $c = 2 \cdot 4 \cdot 0.5 = 4$.

3. Factorise $2x^2 + 5x - 12$

ANSWER $(x + 4)(2x - 3)$

4. The roots of $x^2 - 6x + q = 0$ are equal. Find q and the root.

ANSWER $D = 0$ gives $q = 9$ and $x = 3$.

5. For $x^2 - 5x + 6 = 0$, find $x_1^2 + x_2^2$ without solving.

ANSWER $(x_1 + x_2)^2 - 2x_1x_2 = 25 - 12 = 13$

6. For $x^2 + 3x - 10 = 0$, find $\frac{1}{x_1} + \frac{1}{x_2}$ without solving.

ANSWER $\frac{x_1 + x_2}{x_1x_2} = \frac{-3}{-10} = 0.3$

7. Can $x^2 + x + 3$ be factorised into linear factors? Explain.

ANSWER No — $D = 1 - 12 = -11 < 0$, so there are no real roots.

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 141–146. Use Vieta as a check on every answer.

- Find the sum and product of the roots of $x^2 - 11x + 30 = 0$
- Solve $x^2 - 8x + 15 = 0$ by inspection
- One root of $x^2 + bx - 24 = 0$ is 6. Find the other root and b .
- Factorise $x^2 + 4x - 21$

5. Factorise $2x^2 - 7x + 3$
6. Write a reduced quadratic equation with roots -1 and 8